

Quick Check

Lesson 9: Role of Hormones in Plant Growth and Development: Why Do Maize and Bean Plants Grow Differently?

7 multiple-choice questions • choose A, B, C or D

Sub-Strand 2.2: Growth and Development in Plants | Lesson 9

A farmer pinches off the very top (the shoot tip) of a young bean seedling. What is the most likely result after two weeks?

- A** More side branches grow because the tip no longer holds back the side buds
- B** The plant dies because the tip is cut off
- C** Nothing changes because plants grow only from water and soil
- D** The plant grows taller than before because it has less to feed

In the dry season, healthy seeds in the soil do not germinate even though they are not dead. Which plant hormone helps to keep these seeds dormant?

A Absciscic acid

B Cytokinins

C Gibberellins

D Ethylene

Two bean seedlings were grown for two weeks. The intact seedling was 30 cm tall with 0 side branches. The seedling with its tip pinched off was 18 cm tall with 4 side branches. Which conclusion is best supported by these results?

- A** Removing the tip kills the plant, so it cannot grow taller
- B** Hormones are found only in animals, so the plants differ by chance
- C** Side branches grow only when a plant is shorter than 20 cm
- D** Removing the shoot tip allowed side buds to grow, so the tip normally suppresses branching

Traders in Mombasa put green bananas in a closed sack and they ripen faster. Which explanation is correct?

- A** Absciscic acid is released and speeds up the ripening
- B** Ethylene gas given off by the fruit collects in the sack and speeds up ripening
- C** Auxin builds up in the sack and makes the fruit softer
- D** Cytokinins from the sack make the fruit cells divide faster

What does the term 'apical dominance' mean?

- A** The shoot tip bends towards light because of gibberellins
- B** The side buds always grow faster than the main shoot
- C** The shoot tip makes auxin that holds back the growth of side buds
- D** The root tip controls how fast the shoot grows

Tea farmers in Kericho prune their tea bushes regularly. How does pruning help?

- A** It removes the shoot tips, which reduces apical dominance so side buds grow and the bush becomes bushy
- B** It makes the bush grow taller with one thick stem
- C** It adds abscisic acid to the bush so that it stops growing
- D** It kills the old shoots so that only roots keep growing

A learner revises their Day 1 model of why maize and bean seeds behave differently. Which addition is correct?

- A** Gibberellins promote germination, but the seed also needs water, oxygen and a suitable temperature to respond
- B** Hormones alone decide germination, so water and temperature can be removed from the model
- C** Each hormone does only one job, so no arrows should link hormones to each other
- D** Only animals have hormones, so the model should show only seed structure

All done

Ask your teacher how to submit your answers today (for example, on a paper answer slip).